Project 3

Network Configuration & Security Report:

This project consists of a multi-router enterprise-level network design featuring three internal routers (R1, R2, R3), an ISP edge router, and endpoints distributed across three subnets (Subnet1, Subnet2, Subnet3). The network topology includes:

- **Three LAN segments** hosting PC1, PC2, and PC3
- **Multiple point-to-point serial links** interconnecting R1, R2, and R3
- **A dedicated ISP router** reaching an external Web Server (5.5.5.5)
- **OSPF routing** for dynamic learning of internal and default routes
- **ACLs applied on LAN-facing interfaces** to enforce traffic filtering rules

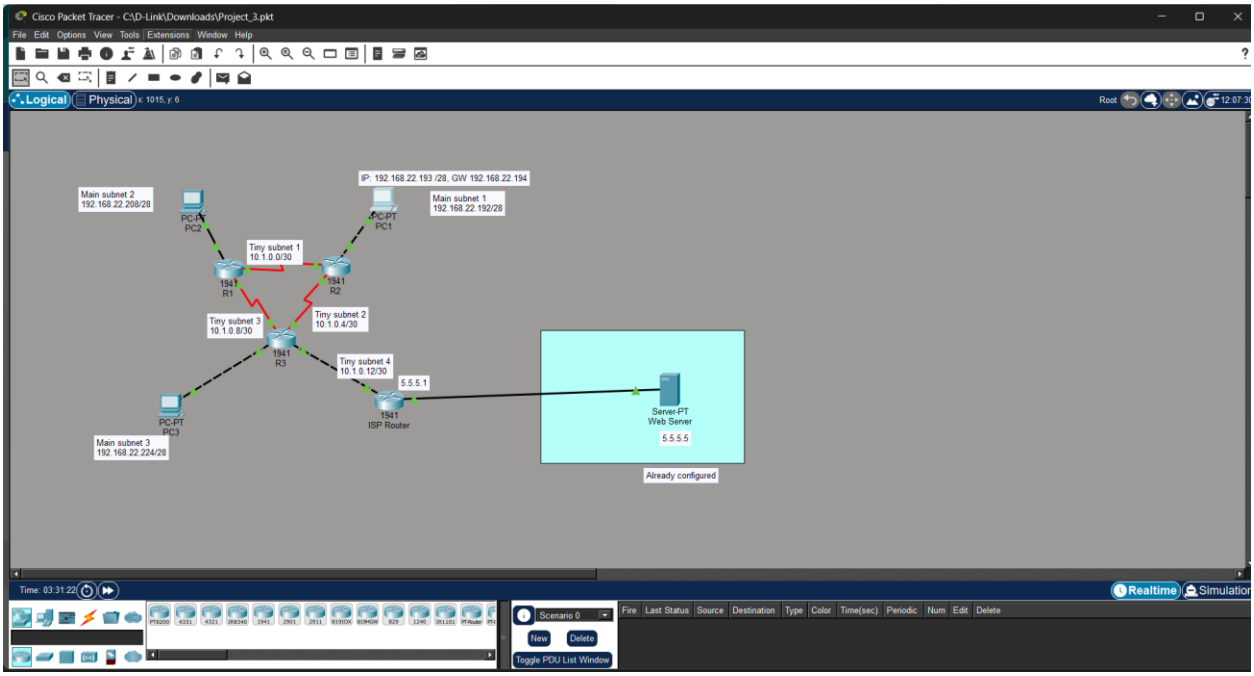
The labeled topology visually identifies:

- LAN subnets (192.168.22.192/28, .208/28, .224/28)
- WAN subnets between routers (10.x.x.x /30)
- Router identifiers and their routing roles
- The centralized position of R3 as the redistribution point for the 0.0.0.0/0 default route

This topology forms the foundation for routing behavior and access control policies implemented in later sections.

Section 1:

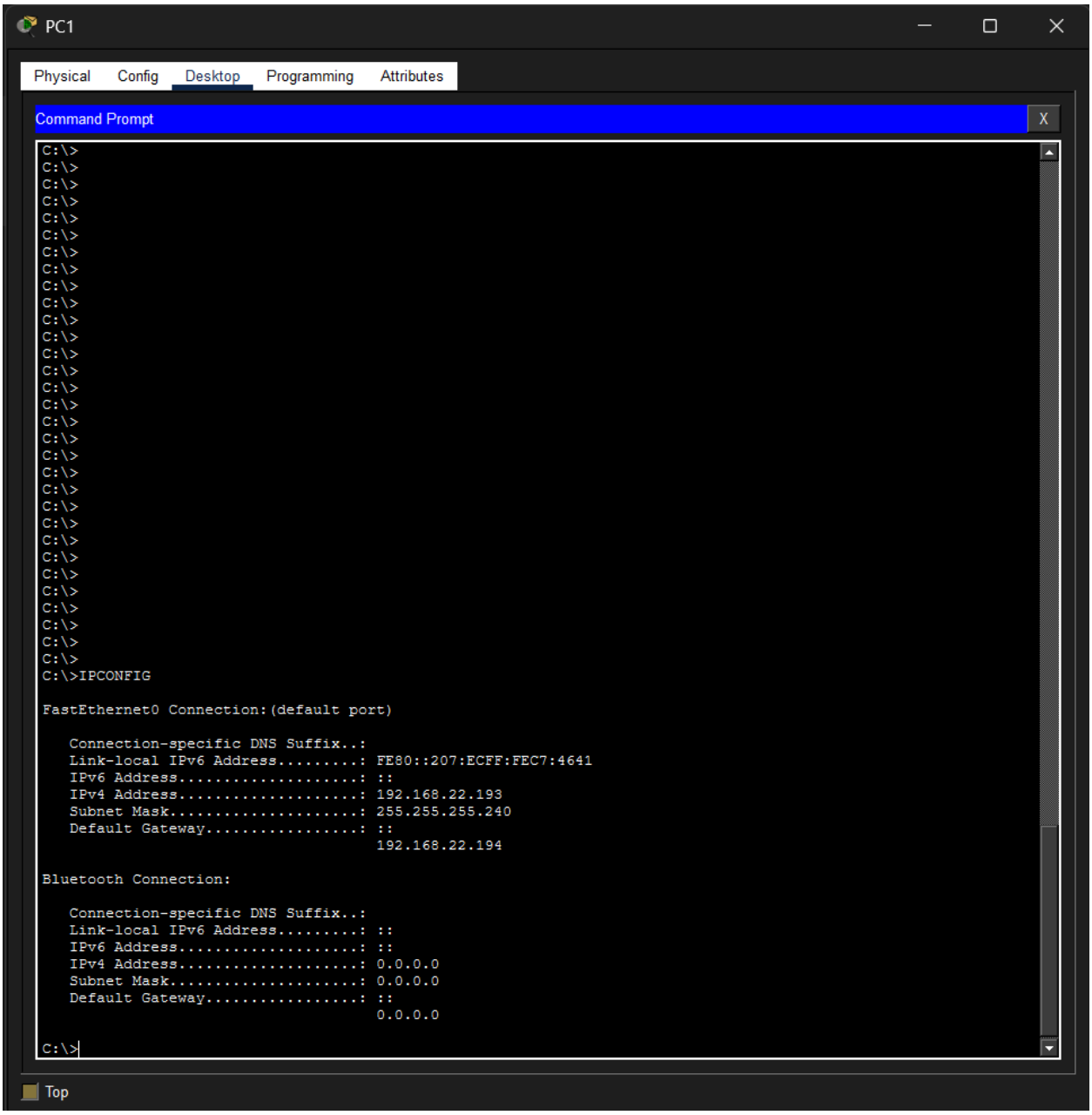
Full Label Network Topology



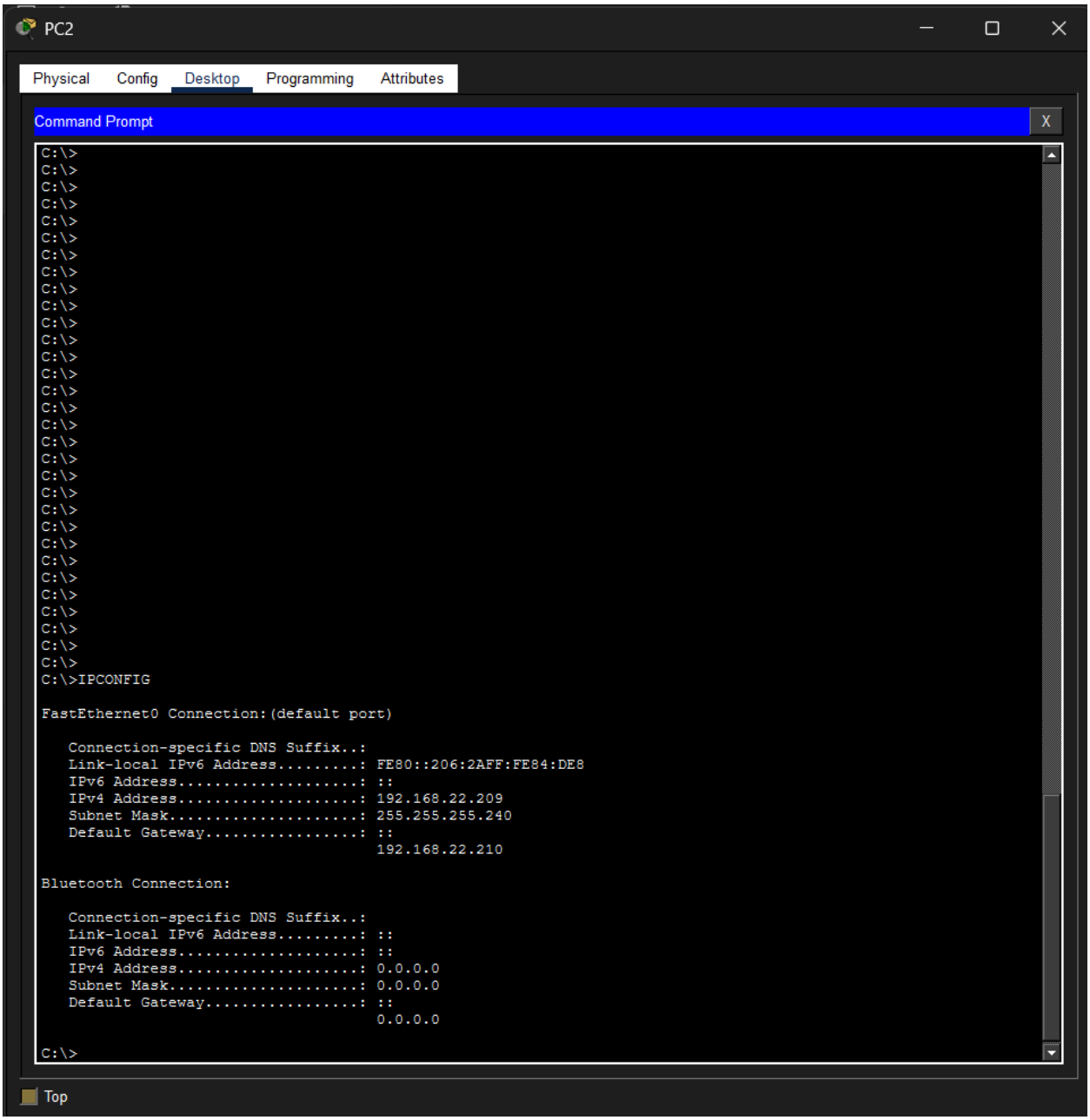
Section 2:

IP Configuration:

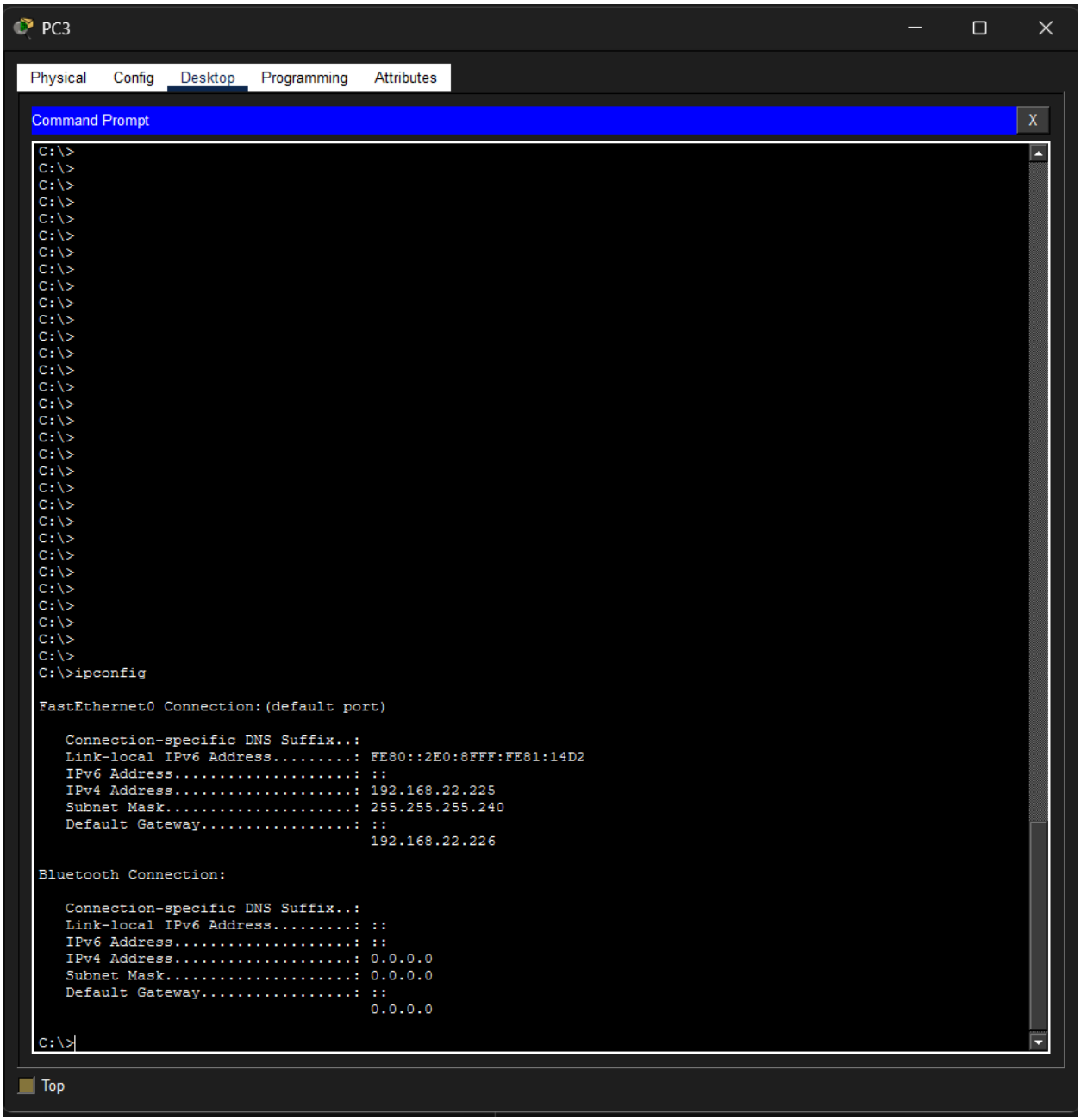
PC1:



PC2:

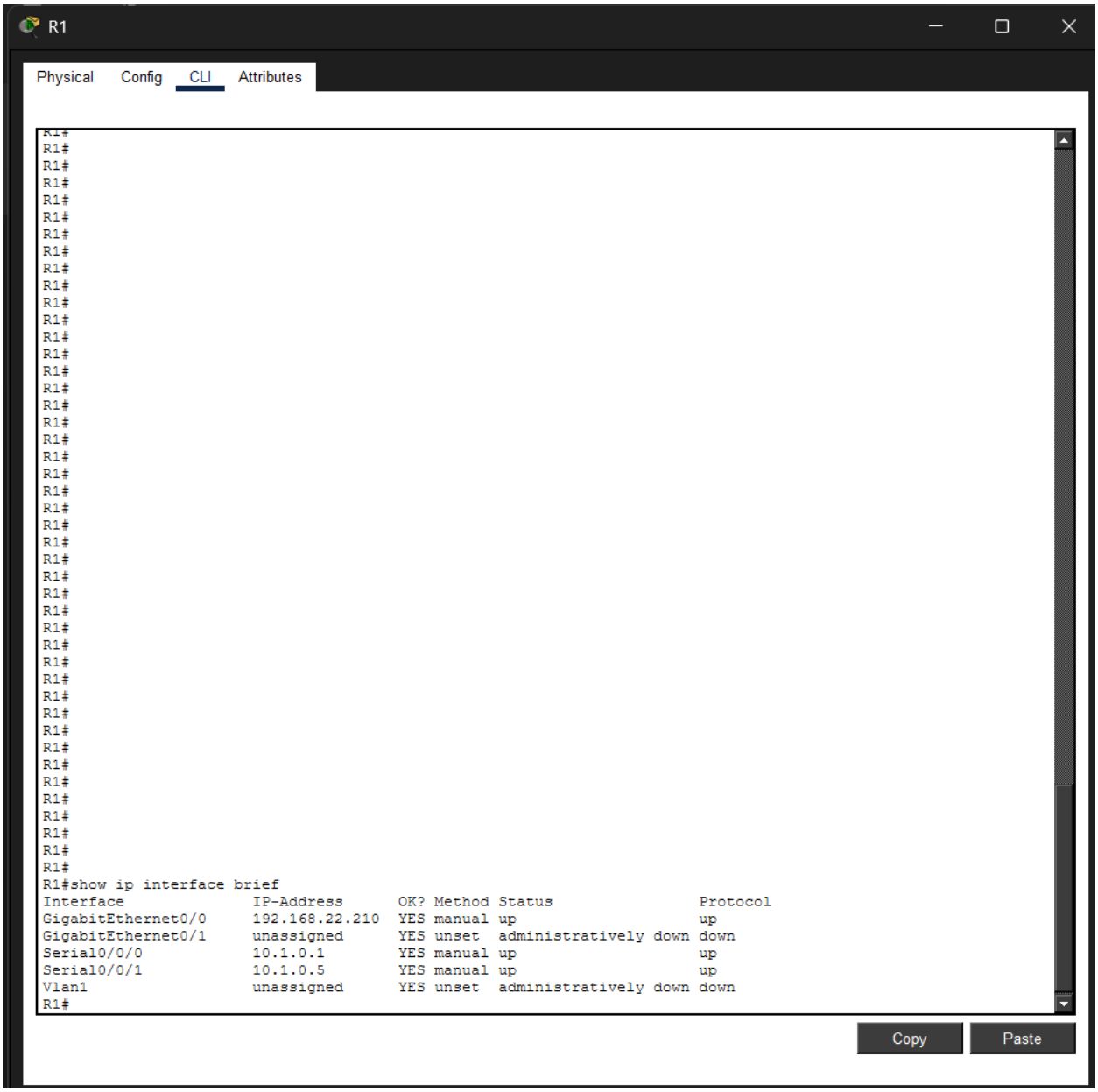

PC2
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
C:\>IPCONFIG
FastEthernet0 Connection:(default port)
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: FE80::206:2AFF:FE84:DE8
IPv6 Address.....: ::
IPv4 Address.....: 192.168.22.209
Subnet Mask.....: 255.255.255.240
Default Gateway.....: ::
192.168.22.210
Bluetooth Connection:
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
0.0.0.0
Top

PC3:


PC3
Physical
Config
Desktop
Programming
Attributes
Command Prompt
C:\>ipconfig
FastEthernet0 Connection:(default port)
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: FE80::2E0:8FFF:FE81:14D2
IPv6 Address.....: ::
IPv4 Address.....: 192.168.22.225
Subnet Mask.....: 255.255.255.240
Default Gateway.....: ::
192.168.22.226
Bluetooth Connection:
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
0.0.0.0
Top

Router Interface Summary:

R1:



R2:

R2

Physical Config CLI Attributes

```
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#





R2 con0 is now available




Press RETURN to get started.


User Access Verification

Username: STUDENT
Password:
% Login invalid

Username: student
Password:

R2>enable
Password:
R2#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.22.194  YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Serial0/0/0            10.1.0.2        YES manual up                    up
Serial0/0/1            10.1.0.9        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R2#
```

Copy Paste

R3:

R3

Physical Config CLI Attributes

```
R3#show access-lists
Extended IP access list SUBNET3_OUT
    10 permit ip 192.168.22.224 0.0.0.15 192.168.22.208 0.0.0.15
    20 permit ip 192.168.22.224 0.0.0.15 192.168.22.192 0.0.0.15
    30 deny ip 192.168.22.224 0.0.0.15 any

R3#









R3 con0 is now available




Press RETURN to get started.


User Access Verification

Username: student
Password:

R3>enable
Password:
R3#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.22.226  YES manual up                    up
GigabitEthernet0/1     10.1.0.13       YES manual up                    up
Serial0/0/0            10.1.0.10       YES manual up                    up
Serial0/0/1            10.1.0.6        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R3#
```

Copy Paste

Top

ISP Router:

ISP Router

Physical Config CLI Attributes

```
Username: student
Password:

ISP_Router>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.1.0.14       YES manual up                    up
GigabitEthernet0/1     5.5.5.1         YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
ISP_Router>




ISP_Router con0 is now available




Press RETURN to get started.


User Access Verification

Username: student
Password:

ISP_Router>enable
Password:
ISP_Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.1.0.14       YES manual up                    up
GigabitEthernet0/1     5.5.5.1         YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
ISP_Router#
```

Copy Paste

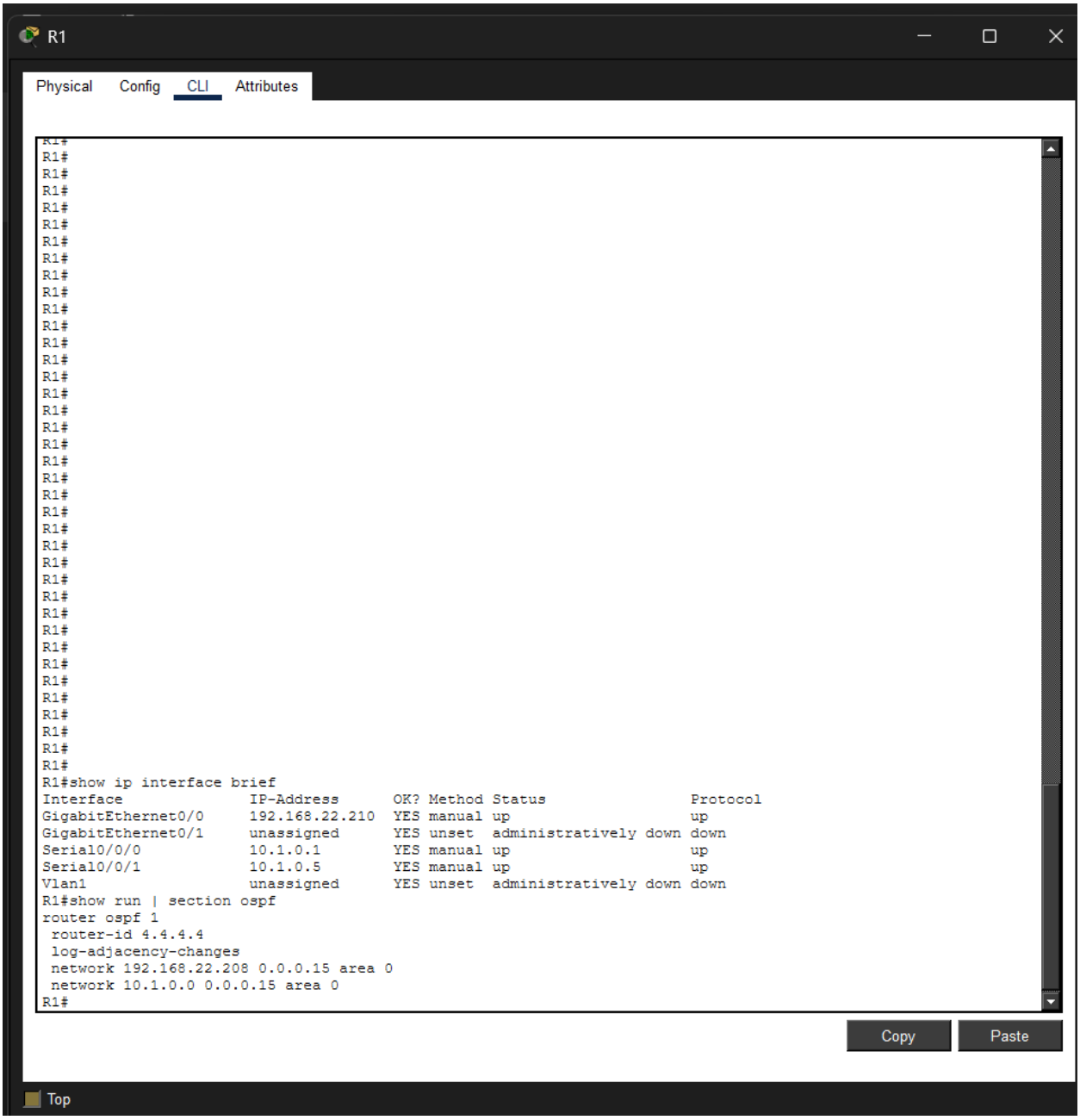
Top

SECTION 3:

OSPF Configurations

Router ospf configuration:

RI:



R2:

R2

Physical Config CLI Attributes

```
R2 con0 is now available



Press RETURN to get started.


User Access Verification

Username: STUDENT
Password:
% Login invalid

Username: student
Password:

R2>enable
Password:
R2#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.22.194  YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Serial0/0/0            10.1.0.2        YES manual up                    up
Serial0/0/1            10.1.0.9        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R2#show run | section ospf
router ospf 1
 router-id 5.5.5.5
 log-adjacency-changes
 network 192.168.22.192 0.0.0.15 area 0
 network 10.1.0.0 0.0.0.15 area 0
R2#
```

Copy Paste

Top

R3:

R3

Physical Config CLI Attributes

```
R3#

R3 con0 is now available

Press RETURN to get started.

User Access Verification

Username: student
Password:

R3>enable
Password:
R3#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.22.226  YES manual up                    up
GigabitEthernet0/1     10.1.0.13       YES manual up                    up
Serial0/0/0            10.1.0.10       YES manual up                    up
Serial0/0/1            10.1.0.6        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R3#show run | section ospf
router ospf 1
 router-id 6.6.6.6
 log-adjacency-changes
 network 192.168.22.224 0.0.0.15 area 0
 network 10.1.0.0 0.0.0.15 area 0
 default-information originate
R3#
```

Copy Paste

Routing Table (OSPF Learned Routes)

R1:

R1

Physical Config CLI Attributes

```
R1#
R1#
R1#
R1#
R1#
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R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.22.210  YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Serial0/0/0            10.1.0.1        YES manual up                    up
Serial0/0/1            10.1.0.5        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R1#show run | section ospf
router ospf 1
 router-id 4.4.4.4
 log-adjacency-changes
 network 192.168.22.208 0.0.0.15 area 0
 network 10.1.0.0 0.0.0.15 area 0
R1#show ip route ospf
     10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
O       10.1.0.8 [110/128] via 10.1.0.2, 04:28:01, Serial0/0/0
                 [110/128] via 10.1.0.6, 04:28:01, Serial0/0/1
O       10.1.0.12 [110/65] via 10.1.0.6, 04:28:01, Serial0/0/1
     192.168.22.0/24 is variably subnetted, 4 subnets, 2 masks
O       192.168.22.192 [110/65] via 10.1.0.2, 04:39:24, Serial0/0/0
O       192.168.22.224 [110/65] via 10.1.0.6, 04:28:01, Serial0/0/1
O*E2 0.0.0.0/0 [110/1] via 10.1.0.6, 04:27:43, Serial0/0/1

R1#
```

Copy Paste

R2:

R2

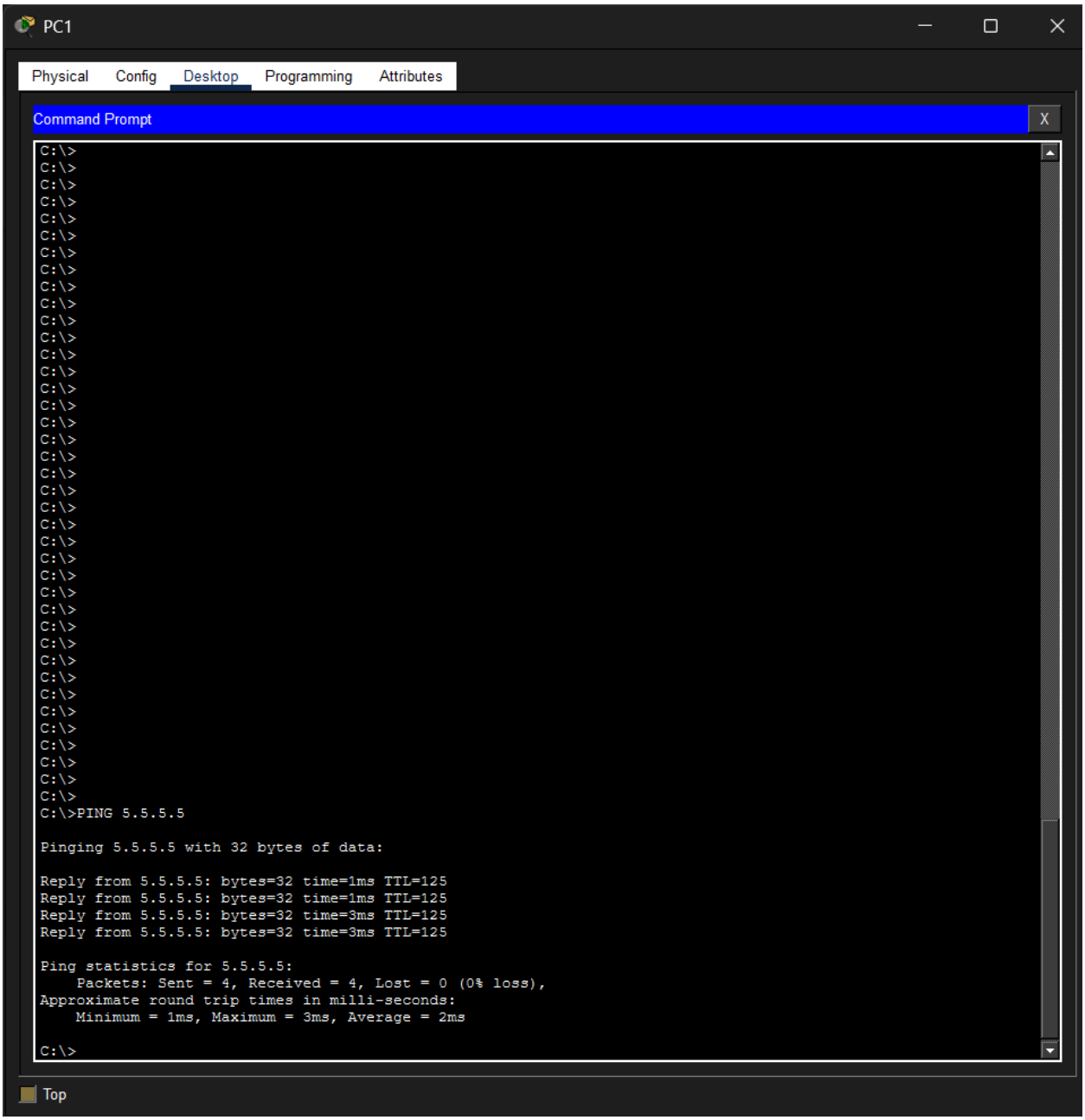
Physical Config CLI Attributes

```
Press RETURN to get started.


User Access Verification

Username: STUDENT
Password:
% Login invalid

Username: student
Password:

R2>enable
Password:
R2#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.22.194  YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Serial0/0/0            10.1.0.2        YES manual up                    up
Serial0/0/1            10.1.0.9        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R2#show run | section ospf
router ospf 1
 router-id 5.5.5.5
 log-adjacency-changes
 network 192.168.22.192 0.0.0.15 area 0
 network 10.1.0.0 0.0.0.15 area 0
R2#show ip route ospf
     10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
O       10.1.0.4 [110/128] via 10.1.0.1, 04:28:43, Serial0/0/0
                 [110/128] via 10.1.0.10, 04:28:43, Serial0/0/1
O       10.1.0.12 [110/65] via 10.1.0.10, 04:28:43, Serial0/0/1
     192.168.22.0/24 is variably subnetted, 4 subnets, 2 masks
O       192.168.22.208 [110/65] via 10.1.0.1, 00:41:39, Serial0/0/0
O       192.168.22.224 [110/65] via 10.1.0.10, 04:28:43, Serial0/0/1
O*E2 0.0.0.0/0 [110/1] via 10.1.0.10, 04:28:25, Serial0/0/1

R2#
```

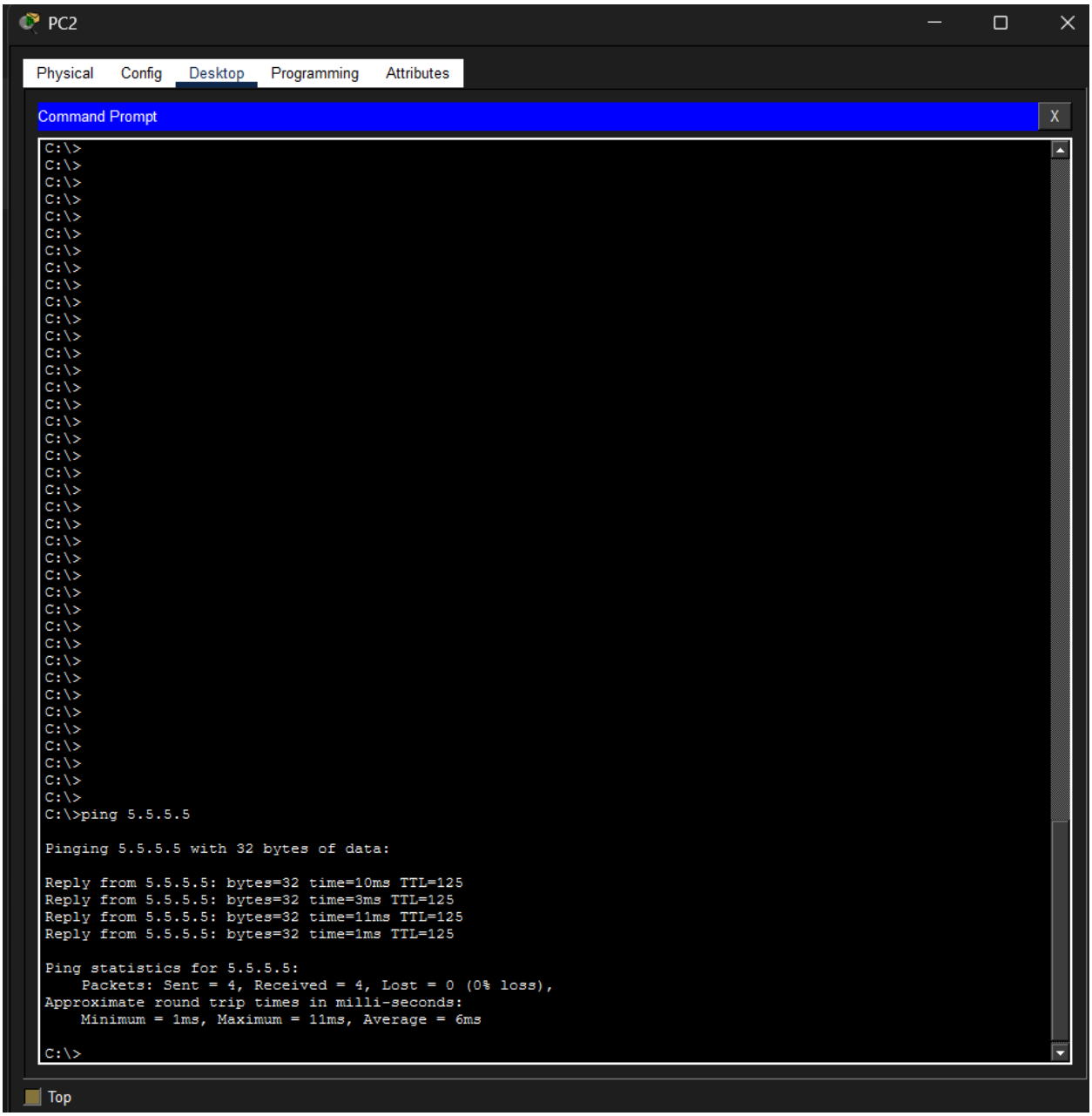
Copy Paste

SECTION 4 — CONNECTIVITY VERIFICATION

PC1 → 5.5.5.5 (success):



PC2 → 5.5.5.5 (success):



PC2 → PC3 (success):

PC2

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
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C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>ping 5.5.5.5

Pinging 5.5.5.5 with 32 bytes of data:

Reply from 5.5.5.5: bytes=32 time=10ms TTL=125
Reply from 5.5.5.5: bytes=32 time=3ms TTL=125
Reply from 5.5.5.5: bytes=32 time=11ms TTL=125
Reply from 5.5.5.5: bytes=32 time=1ms TTL=125

Ping statistics for 5.5.5.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 11ms, Average = 6ms

C:\>ping 192.168.22.225

Pinging 192.168.22.225 with 32 bytes of data:

Reply from 192.168.22.225: bytes=32 time=2ms TTL=126
Reply from 192.168.22.225: bytes=32 time=1ms TTL=126
Reply from 192.168.22.225: bytes=32 time=1ms TTL=126
Reply from 192.168.22.225: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.22.225:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>
```
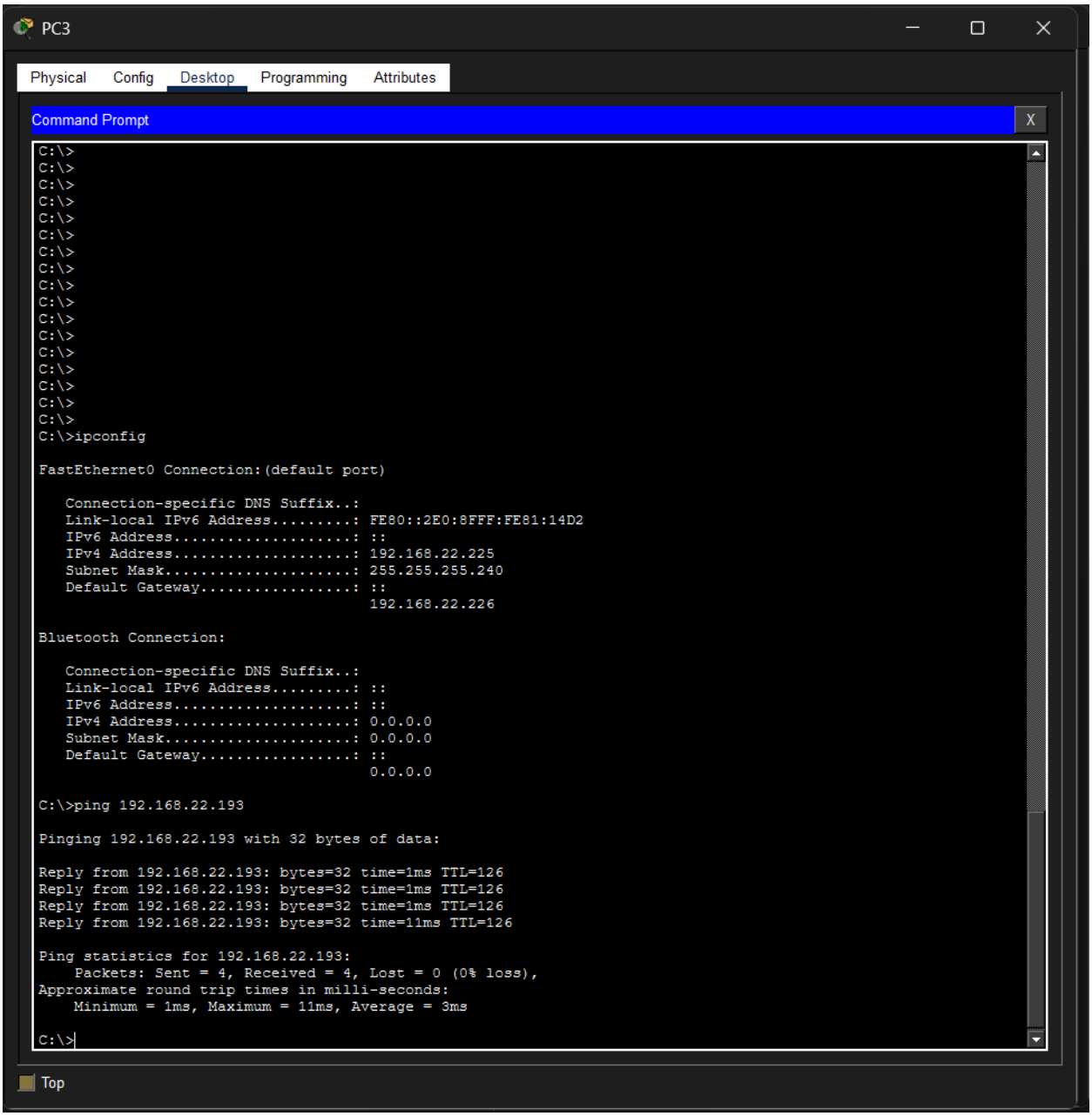

Top

PC3 → PC1 (success):



PC3 → PC2 (success):

PC3

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>
C:\>
C:\>
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::2E0:8FFF:FE81:14D2
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.22.225
   Subnet Mask.....................: 255.255.255.240
   Default Gateway.................: ::
                                     192.168.22.226

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 192.168.22.193

Pinging 192.168.22.193 with 32 bytes of data:

Reply from 192.168.22.193: bytes=32 time=1ms TTL=126
Reply from 192.168.22.193: bytes=32 time=1ms TTL=126
Reply from 192.168.22.193: bytes=32 time=1ms TTL=126
Reply from 192.168.22.193: bytes=32 time=11ms TTL=126

Ping statistics for 192.168.22.193:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 11ms, Average = 3ms

C:\>ping 192.168.22.209

Pinging 192.168.22.209 with 32 bytes of data:

Reply from 192.168.22.209: bytes=32 time=1ms TTL=126
Reply from 192.168.22.209: bytes=32 time=3ms TTL=126
Reply from 192.168.22.209: bytes=32 time=3ms TTL=126
Reply from 192.168.22.209: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.22.209:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\>
```

Top

Blocked Pings

PC1 → PC2 (blocked):

PC1

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>PING 5.5.5.5

Pinging 5.5.5.5 with 32 bytes of data:

Reply from 5.5.5.5: bytes=32 time=1ms TTL=125
Reply from 5.5.5.5: bytes=32 time=1ms TTL=125
Reply from 5.5.5.5: bytes=32 time=3ms TTL=125
Reply from 5.5.5.5: bytes=32 time=3ms TTL=125

Ping statistics for 5.5.5.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::207:ECFF:FEC7:4641
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.22.193
   Subnet Mask.....................: 255.255.255.240
   Default Gateway.................: ::
                                     192.168.22.194

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 192.168.22.209

Pinging 192.168.22.209 with 32 bytes of data:

Reply from 192.168.22.194: Destination host unreachable.
Reply from 192.168.22.194: Destination host unreachable.
Reply from 192.168.22.194: Destination host unreachable.
Reply from 192.168.22.194: Destination host unreachable.

Ping statistics for 192.168.22.209:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Top

PC2 → PC1 (blocked):

PC2

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Ping statistics for 5.5.5.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 11ms, Average = 6ms

C:\>ping 192.168.22.225

Pinging 192.168.22.225 with 32 bytes of data:

Reply from 192.168.22.225: bytes=32 time=2ms TTL=126
Reply from 192.168.22.225: bytes=32 time=1ms TTL=126
Reply from 192.168.22.225: bytes=32 time=1ms TTL=126
Reply from 192.168.22.225: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.22.225:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::206:2AFF:FE84:DE8
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.22.209
   Subnet Mask.....................: 255.255.255.240
   Default Gateway.................: ::
                                     192.168.22.210

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 192.168.22.193

Pinging 192.168.22.193 with 32 bytes of data:

Reply from 192.168.22.210: Destination host unreachable.
Reply from 192.168.22.210: Destination host unreachable.
Reply from 192.168.22.210: Destination host unreachable.
Reply from 192.168.22.210: Destination host unreachable.

Ping statistics for 192.168.22.193:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Top

PC3 → 5.5.5.5 (blocked):

PC3

Physical Config Desktop Programming Attributes

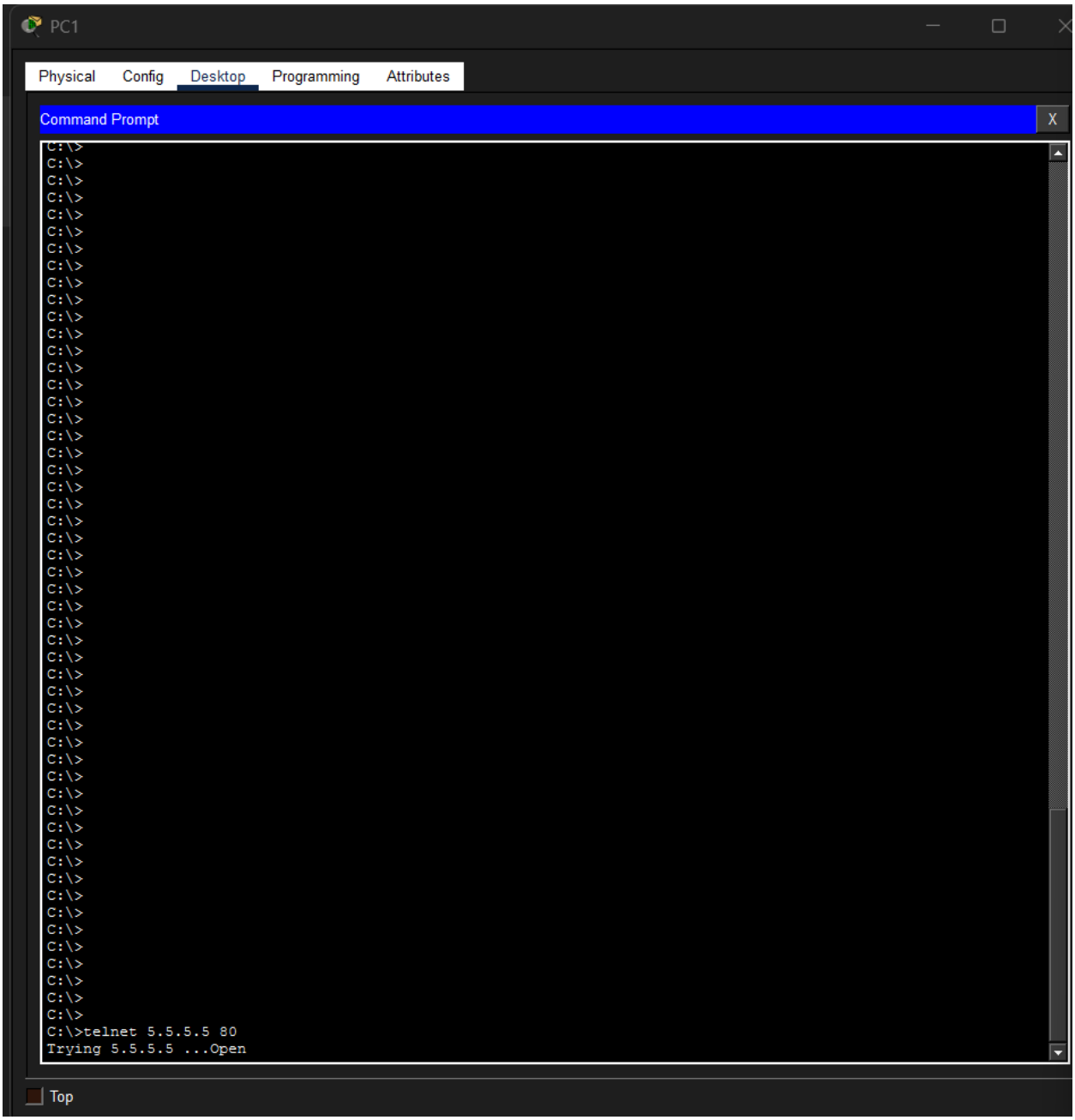
Command Prompt

```
   Default Gateway.................: ::
                                     192.168.22.226

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 192.168.22.193

Pinging 192.168.22.193 with 32 bytes of data:

Reply from 192.168.22.193: bytes=32 time=1ms TTL=126
Reply from 192.168.22.193: bytes=32 time=1ms TTL=126
Reply from 192.168.22.193: bytes=32 time=1ms TTL=126
Reply from 192.168.22.193: bytes=32 time=11ms TTL=126

Ping statistics for 192.168.22.193:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 11ms, Average = 3ms

C:\>ping 192.168.22.209

Pinging 192.168.22.209 with 32 bytes of data:

Reply from 192.168.22.209: bytes=32 time=1ms TTL=126
Reply from 192.168.22.209: bytes=32 time=3ms TTL=126
Reply from 192.168.22.209: bytes=32 time=3ms TTL=126
Reply from 192.168.22.209: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.22.209:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\>ping 5.5.5.5

Pinging 5.5.5.5 with 32 bytes of data:

Reply from 192.168.22.226: Destination host unreachable.
Reply from 192.168.22.226: Destination host unreachable.
Reply from 192.168.22.226: Destination host unreachable.
Reply from 192.168.22.226: Destination host unreachable.

Ping statistics for 5.5.5.5:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Top

Web Access Tests

PC1:



PC2:

PC2

Physical Config Desktop Programming Attributes

Command Prompt

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 11ms, Average = 6ms

C:\>ping 192.168.22.225

Pinging 192.168.22.225 with 32 bytes of data:

Reply from 192.168.22.225: bytes=32 time=2ms TTL=126
Reply from 192.168.22.225: bytes=32 time=1ms TTL=126
Reply from 192.168.22.225: bytes=32 time=1ms TTL=126
Reply from 192.168.22.225: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.22.225:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::206:2AFF:FE84:DE8
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.22.209
   Subnet Mask.....................: 255.255.255.240
   Default Gateway.................: ::
                                     192.168.22.210

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 192.168.22.193

Pinging 192.168.22.193 with 32 bytes of data:

Reply from 192.168.22.210: Destination host unreachable.
Reply from 192.168.22.210: Destination host unreachable.
Reply from 192.168.22.210: Destination host unreachable.
Reply from 192.168.22.210: Destination host unreachable.

Ping statistics for 192.168.22.193:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>telnet 5.5.5.5 80
Trying 5.5.5.5 ...Open
```

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SECTION 5 — ACL CONFIGURATION & MATCHES:

R1:

R1

Physical Config CLI Attributes

```
R1#
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R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.22.210  YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Serial0/0/0            10.1.0.1        YES manual up                    up
Serial0/0/1            10.1.0.5        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R1#show run | section ospf
router ospf 1
 router-id 4.4.4.4
 log-adjacency-changes
 network 192.168.22.208 0.0.0.15 area 0
 network 10.1.0.0 0.0.0.15 area 0
R1#show ip route ospf
     10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
O       10.1.0.8 [110/128] via 10.1.0.2, 04:28:01, Serial0/0/0
                 [110/128] via 10.1.0.6, 04:28:01, Serial0/0/1
O       10.1.0.12 [110/65] via 10.1.0.6, 04:28:01, Serial0/0/1
     192.168.22.0/24 is variably subnetted, 4 subnets, 2 masks
O       192.168.22.192 [110/65] via 10.1.0.2, 04:39:24, Serial0/0/0
O       192.168.22.224 [110/65] via 10.1.0.6, 04:28:01, Serial0/0/1
O*E2 0.0.0.0/0 [110/1] via 10.1.0.6, 04:27:43, Serial0/0/1

R1#show run | section access-list
ip access-list extended SUBNET2_OUT
 permit icmp host 192.168.22.209 host 5.5.5.5
 permit tcp host 192.168.22.209 host 5.5.5.5 eq www
 permit ip 192.168.22.208 0.0.0.15 192.168.22.224 0.0.0.15
 deny ip 192.168.22.208 0.0.0.15 any
R1#
```

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R2:

R2

Physical Config CLI Attributes

```
Password:
% Login invalid

Username: student
Password:

R2>enable
Password:
R2#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.22.194  YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Serial0/0/0            10.1.0.2        YES manual up                    up
Serial0/0/1            10.1.0.9        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R2#show run | section ospf
router ospf 1
 router-id 5.5.5.5
 log-adjacency-changes
 network 192.168.22.192 0.0.0.15 area 0
 network 10.1.0.0 0.0.0.15 area 0
R2#show ip route ospf
     10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
O       10.1.0.4 [110/128] via 10.1.0.1, 04:28:43, Serial0/0/0
                 [110/128] via 10.1.0.10, 04:28:43, Serial0/0/1
O       10.1.0.12 [110/65] via 10.1.0.10, 04:28:43, Serial0/0/1
     192.168.22.0/24 is variably subnetted, 4 subnets, 2 masks
O       192.168.22.208 [110/65] via 10.1.0.1, 00:41:39, Serial0/0/0
O       192.168.22.224 [110/65] via 10.1.0.10, 04:28:43, Serial0/0/1
O*E2 0.0.0.0/0 [110/1] via 10.1.0.10, 04:28:25, Serial0/0/1

R2#show run | section access-list
ip access-list extended SUBNET1_OUT
 permit icmp 192.168.22.192 0.0.0.15 host 5.5.5.5
 permit tcp 192.168.22.192 0.0.0.15 host 5.5.5.5 eq www
 permit ip 192.168.22.192 0.0.0.15 192.168.22.224 0.0.0.15
 deny ip 192.168.22.192 0.0.0.15 any
```

R3:

R3

Physical Config CLI Attributes

```
R3 con0 is now available



Press RETURN to get started.



User Access Verification

Username: student
Password:

R3>enable
Password:
R3#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.22.226  YES manual up                    up
GigabitEthernet0/1     10.1.0.13       YES manual up                    up
Serial0/0/0            10.1.0.10       YES manual up                    up
Serial0/0/1            10.1.0.6        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R3#show run | section ospf
router ospf 1
 router-id 6.6.6.6
 log-adjacency-changes
 network 192.168.22.224 0.0.0.15 area 0
 network 10.1.0.0 0.0.0.15 area 0
 default-information originate
R3#show run | section access-list
ip access-list extended SUBNET3_OUT
 permit ip 192.168.22.224 0.0.0.15 192.168.22.208 0.0.0.15
 permit ip 192.168.22.224 0.0.0.15 192.168.22.192 0.0.0.15
 deny ip 192.168.22.224 0.0.0.15 any
R3#
```

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ACL Matches:

R1:

R1

Physical Config CLI Attributes

```
R1#
R1#
R1#
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R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.22.210  YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Serial0/0/0            10.1.0.1        YES manual up                    up
Serial0/0/1            10.1.0.5        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R1#show run | section ospf
router ospf 1
 router-id 4.4.4.4
 log-adjacency-changes
 network 192.168.22.208 0.0.0.15 area 0
 network 10.1.0.0 0.0.0.15 area 0
R1#show ip route ospf
     10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
O       10.1.0.8 [110/128] via 10.1.0.2, 04:28:01, Serial0/0/0
                 [110/128] via 10.1.0.6, 04:28:01, Serial0/0/1
O       10.1.0.12 [110/65] via 10.1.0.6, 04:28:01, Serial0/0/1
     192.168.22.0/24 is variably subnetted, 4 subnets, 2 masks
O       192.168.22.192 [110/65] via 10.1.0.2, 04:39:24, Serial0/0/0
O       192.168.22.224 [110/65] via 10.1.0.6, 04:28:01, Serial0/0/1
O*E2 0.0.0.0/0 [110/1] via 10.1.0.6, 04:27:43, Serial0/0/1

R1#show run | section access-list
ip access-list extended SUBNET2_OUT
 permit icmp host 192.168.22.209 host 5.5.5.5
 permit tcp host 192.168.22.209 host 5.5.5.5 eq www
 permit ip 192.168.22.208 0.0.0.15 192.168.22.224 0.0.0.15
 deny ip 192.168.22.208 0.0.0.15 any
R1#show access-lists
Extended IP access list SUBNET2_OUT
    10 permit icmp host 192.168.22.209 host 5.5.5.5 (8 match(es))
    20 permit tcp host 192.168.22.209 host 5.5.5.5 eq www (2 match(es))
    30 permit ip 192.168.22.208 0.0.0.15 192.168.22.224 0.0.0.15 (15 match(es))
    40 deny ip 192.168.22.208 0.0.0.15 any (4 match(es))

R1#
```

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R2:

R2

Physical Config CLI Attributes

```
Password:
R2#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.22.194  YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Serial0/0/0            10.1.0.2        YES manual up                    up
Serial0/0/1            10.1.0.9        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R2#show run | section ospf
router ospf 1
 router-id 5.5.5.5
 log-adjacency-changes
 network 192.168.22.192 0.0.0.15 area 0
 network 10.1.0.0 0.0.0.15 area 0
R2#show ip route ospf
     10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
O       10.1.0.4 [110/128] via 10.1.0.1, 04:28:43, Serial0/0/0
                 [110/128] via 10.1.0.10, 04:28:43, Serial0/0/1
O       10.1.0.12 [110/65] via 10.1.0.10, 04:28:43, Serial0/0/1
     192.168.22.0/24 is variably subnetted, 4 subnets, 2 masks
O       192.168.22.208 [110/65] via 10.1.0.1, 00:41:39, Serial0/0/0
O       192.168.22.224 [110/65] via 10.1.0.10, 04:28:43, Serial0/0/1
O*E2 0.0.0.0/0 [110/1] via 10.1.0.10, 04:28:25, Serial0/0/1

R2#show run | section access-list
ip access-list extended SUBNET1_OUT
 permit icmp 192.168.22.192 0.0.0.15 host 5.5.5.5
 permit tcp 192.168.22.192 0.0.0.15 host 5.5.5.5 eq www
 permit ip 192.168.22.192 0.0.0.15 192.168.22.224 0.0.0.15
 deny ip 192.168.22.192 0.0.0.15 any
R2#show run interface GigabitEthernet0/0
            ^
% Invalid input detected at '^' marker.

R2#show run interface GigabitEthernet0/0
            ^
% Invalid input detected at '^' marker.

R2#show run | section access-list
ip access-list extended SUBNET1_OUT
 permit icmp 192.168.22.192 0.0.0.15 host 5.5.5.5
 permit tcp 192.168.22.192 0.0.0.15 host 5.5.5.5 eq www
 permit ip 192.168.22.192 0.0.0.15 192.168.22.224 0.0.0.15
 deny ip 192.168.22.192 0.0.0.15 any
R2#show access-lists
Extended IP access list SUBNET1_OUT
    10 permit icmp 192.168.22.192 0.0.0.15 host 5.5.5.5 (8 match(es))
    20 permit tcp 192.168.22.192 0.0.0.15 host 5.5.5.5 eq www (10 match(es))
    30 permit ip 192.168.22.192 0.0.0.15 192.168.22.224 0.0.0.15 (12 match(es))
    40 deny ip 192.168.22.192 0.0.0.15 any (24 match(es))

R2#
```

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R3:

R3

Physical Config CLI Attributes

```
Press RETURN to get started.


User Access Verification

Username: student
Password:

R3>enable
Password:
R3#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.22.226  YES manual up                    up
GigabitEthernet0/1     10.1.0.13       YES manual up                    up
Serial0/0/0            10.1.0.10       YES manual up                    up
Serial0/0/1            10.1.0.6        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R3#show run | section ospf
router ospf 1
 router-id 6.6.6.6
 log-adjacency-changes
 network 192.168.22.224 0.0.0.15 area 0
 network 10.1.0.0 0.0.0.15 area 0
 default-information originate
R3#show run | section access-list
ip access-list extended SUBNET3_OUT
 permit ip 192.168.22.224 0.0.0.15 192.168.22.208 0.0.0.15
 permit ip 192.168.22.224 0.0.0.15 192.168.22.192 0.0.0.15
 deny ip 192.168.22.224 0.0.0.15 any
R3#show access-lists
Extended IP access list SUBNET3_OUT
    10 permit ip 192.168.22.224 0.0.0.15 192.168.22.208 0.0.0.15 (24 match(es))
    20 permit ip 192.168.22.224 0.0.0.15 192.168.22.192 0.0.0.15 (12 match(es))
    30 deny ip 192.168.22.224 0.0.0.15 any (16 match(es))

R3#
```

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SECTION 6 — FINAL OVERVIEW

R1

Physical Config CLI Attributes

```
router ospf 1
 router-id 4.4.4.4
 log-adjacency-changes
 network 192.168.22.208 0.0.0.15 area 0
 network 10.1.0.0 0.0.0.15 area 0
R1#show ip route ospf
     10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
O       10.1.0.8 [110/128] via 10.1.0.2, 04:28:01, Serial0/0/0
                 [110/128] via 10.1.0.6, 04:28:01, Serial0/0/1
O       10.1.0.12 [110/65] via 10.1.0.6, 04:28:01, Serial0/0/1
     192.168.22.0/24 is variably subnetted, 4 subnets, 2 masks
O       192.168.22.192 [110/65] via 10.1.0.2, 04:39:24, Serial0/0/0
O       192.168.22.224 [110/65] via 10.1.0.6, 04:28:01, Serial0/0/1
O*E2 0.0.0.0/0 [110/1] via 10.1.0.6, 04:27:43, Serial0/0/1

R1#show run | section access-list
ip access-list extended SUBNET2_OUT
 permit icmp host 192.168.22.209 host 5.5.5.5
 permit tcp host 192.168.22.209 host 5.5.5.5 eq www
 permit ip 192.168.22.208 0.0.0.15 192.168.22.224 0.0.0.15
 deny ip 192.168.22.208 0.0.0.15 any
R1#show access-lists
Extended IP access list SUBNET2_OUT
    10 permit icmp host 192.168.22.209 host 5.5.5.5 (8 match(es))
    20 permit tcp host 192.168.22.209 host 5.5.5.5 eq www (2 match(es))
    30 permit ip 192.168.22.208 0.0.0.15 192.168.22.224 0.0.0.15 (15 match(es))
    40 deny ip 192.168.22.208 0.0.0.15 any (4 match(es))

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.1.0.6 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C       10.1.0.0/30 is directly connected, Serial0/0/0
L       10.1.0.1/32 is directly connected, Serial0/0/0
C       10.1.0.4/30 is directly connected, Serial0/0/1
L       10.1.0.5/32 is directly connected, Serial0/0/1
O       10.1.0.8/30 [110/128] via 10.1.0.2, 04:51:28, Serial0/0/0
                    [110/128] via 10.1.0.6, 04:51:28, Serial0/0/1
O       10.1.0.12/30 [110/65] via 10.1.0.6, 04:51:28, Serial0/0/1
     192.168.22.0/24 is variably subnetted, 4 subnets, 2 masks
O       192.168.22.192/28 [110/65] via 10.1.0.2, 05:02:51, Serial0/0/0
C       192.168.22.208/28 is directly connected, GigabitEthernet0/0
L       192.168.22.210/32 is directly connected, GigabitEthernet0/0
 --More--
```

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CONCLUSION:

This project demonstrates a fully functional enterprise-style configuration integrating:

- **Hierarchical subnetting using /28 and /30 networks**
- **OSPF dynamic routing with default route injection**
- **Granular ACL enforcement tailored to each subnet**
- **Successful verification through extensive ping and web access testing**

All configurations, ACL logic, and verification steps meet the project requirements. The network behaves predictably, routing is efficient, and security policies effectively control inter-subnet communication and external access.